FAMOTIDINE

Renal dose adjustment

Few interactions

CHEMICAL NAME 3-[({2-[(diaminomethylidene)amino]-1,3-thiazol-4-yl}methyl)sulfanyl]-N′-sulfamoylpropanimidamide

CLASS Histamine H2-receptor antagonist

TRADE NAMES Pepcid · Pepcidine · Famotin · Antodine

THERAPEUTIC USES / INDICATIONS

- Duodenal and gastric ulcer; GORD.
- Dyspepsia, heartburn (OTC).
- Zollinger–Ellison syndrome.
- Pre-anaesthetic (aspiration prophylaxis); urticaria (adjunct with H1).

PHARMACODYNAMICS

- Competitively blocks **H2 receptors** on parietal cells → ↓ histamine-stimulated (especially nocturnal) acid secretion.
- Less potent than PPIs.

ADMINISTRATION

- **PO** at bedtime or twice daily, with or without food.
- **IV** slowly over 2 min or infusion over 15–30 min.

DOSAGES

- **Ulcer:** 40 mg at night (or 20 mg twice daily) for 4–8 weeks.
- **GORD:** 20–40 mg twice daily.
- **Heartburn (OTC):** 10–20 mg up to twice daily.
- **IV:** 20 mg every 12 h.
- **CrCl <50:** halve dose or give every 36–48 h.

PHARMACOKINETICS

- **Onset:** PO 1 h; IV 30 min. **Duration:** 10–12 h.
- Minimal liver metabolism (no CYP inhibition).
- Excreted unchanged in urine. **t½:** 2.5–3.5 h.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Headache, dizziness.
- Constipation or diarrhoea.

ADVERSE EFFECTS (SERIOUS)

- Confusion, agitation (elderly, renal impairment).
- QT prolongation in renal failure.
- Thrombocytopenia (rare).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity to H2 antagonists.
- **Caution:** renal impairment, elderly.

KEY INTERACTIONS

- **Atazanavir, ketoconazole, itraconazole:** ↓ absorption.
- Unlike cimetidine — no significant CYP interactions.

NURSING MANAGEMENT

- Adjust dose for renal function.
- Monitor mental status in elderly.
- Assess epigastric pain, GI bleeding signs.

PATIENT EDUCATION

- Take at bedtime for ulcers.
- Avoid smoking, alcohol, NSAIDs.
- Report black stools, vomiting blood.

KEY POINTS

- H2 blocker; less potent than PPIs.
- Few drug interactions.
- Renal dose adjustment.
- Confusion in elderly.